

## AI and ML services in the Fintech industry

We live in a data driven world, where it is impossible to secure applications or online services manually. AI based solutions are the need of the hour for anomaly detection in transactions and user services.

AI based solutions in Fintech enable greater security, improved customer experience and return on investment, and faster time to market. A few AI driven solutions being used in the Fintech industry are listed below.

### 1. **Gamification in investment banking:**

Platforms like Betfair.com and WikiHow use AI based algorithms for intelligent gamification to predict prices for stocks based on various parameters like history, trending conditions, market need and stock movement. This is intelligent gambling in a three-minute cycle to give quotes for trading options.

### 2. **Finance-as-a-Service:** This is a new age service where cloud providers like Azure release Microsoft Cloud for financial services. Amazon FinSpace is another example. Open banking APIs and SaaS solutions for Fintech services (like Mambu) help to quickly get into digital banking services.

### 3. **Fraud Prevention-as-a-Service:** SaaS platforms like Sardine.ai give solutions for fraud detection using AI to analyse credit card transactions, customer credit history, and online shopping related frauds.

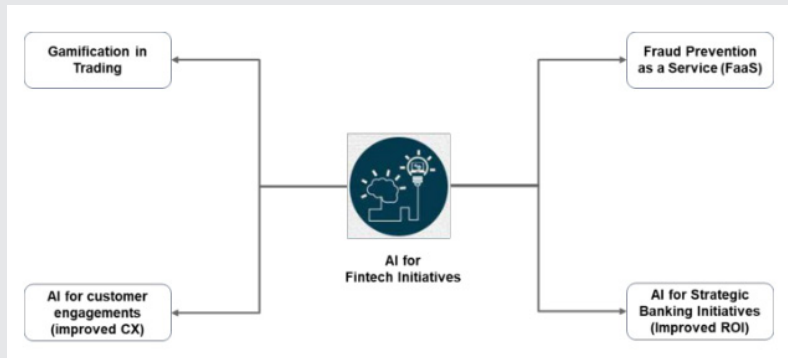


Figure 3: AI for Fintech use cases

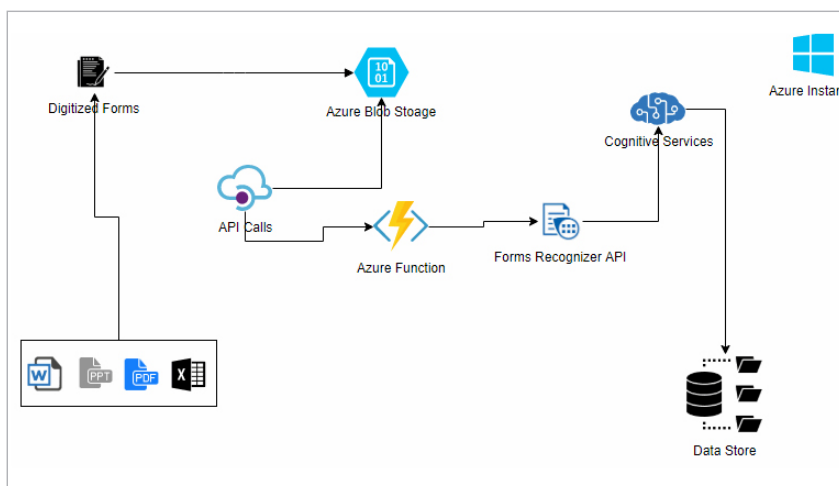


Figure 4: Azure Forms Recognizer solution for data extraction from forms

connection to real systems via REST API). The *Manage* section is for the system behind the scenes. It encompasses the computation clusters and instances, data stores on which the data sets are stored, and the integrations into other systems. It is an invisible layer.

Azure Machine Learning Studio creates a machine learning model in seven steps:

- Data gathering:** Gathering of data starts the process, as data correction leads to perfect prediction models. MAML Studio provides a number of sample data sets, and additional collections can be downloaded from [archives.ics.uci.edu/ml/dataset.html](https://archives.ics.uci.edu/ml/dataset.html).
- Data preview:** After uploading the data, the data is previewed to validate the data input.

- Modules integration:** After viewing, modules are added by a simple drag and drop from the panel on the left.
- Data preprocessing:** This involves adjusting the available data as per nodes. Statistical data is computed from the available data.
- Defining features:** In this step, the features of the data set that are most relevant to the result for prediction are determined.
- Applying algorithms:** The algorithm is applied to train the model.
- Predicting:** The prediction is tested and data is visualised in this step.

## Azure Cognitive Services

Azure Cognitive Services are cloud based services with APIs (application programming interfaces) and SDKs (software development kits) which allow developers to integrate cognitive intelligence into applications even if they don't have a good knowledge of AI. With Azure Cognitive Services, various features like face recognition,